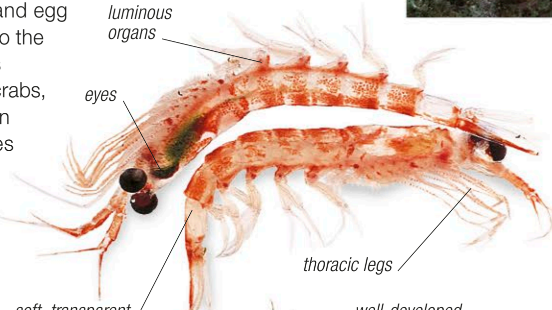
Class Malacostraca

Malacostracans

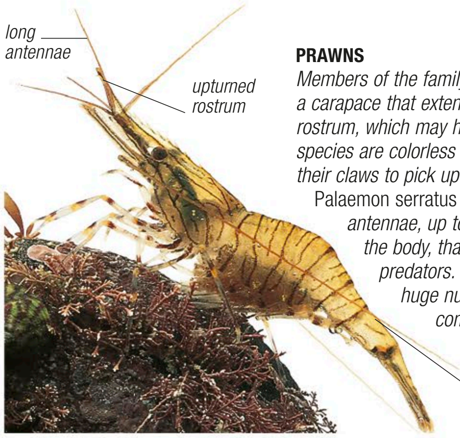
Occurrence 38,000 spp. worldwide; in marine intertidal to abyssal zones, freshwater, terrestrial habitats

The world's largest arthropod, the Japanese spider crab (*Macrocheira kaempferi*) belongs to this large and diverse group of crustaceans, which comprises 18 orders and 606 families. It has a legspan of 13 ft (4 m)—in marked contrast to some other malacostracans, which are less than 1⁄32 in (1 mm) long. Despite this vast difference in size, malacostracans have many features in common. They are often brightly colored, and have tough exoskeletons strengthened with calcium carbonate. Their carapace—if present—acts as a gill chamber, and never covers the abdomen. They often have stalked eyes, and their antennae are prominent. The thorax is made up of 8 segments, while the abdomen has 6 segments and a flattened tail fan (telson) that can be moved rapidly in order to propel the animal away from predators. In many species, the first 1–3 thoracic appendages have become modified into maxillipeds, which are used in feeding; the remaining are used for locomotion. The abdominal appendages help in swimming, burrowing, mating, and egg brooding. Most species belong to the order Decapoda, which contains all the familiar shrimps, crayfish, crabs, and lobsters. They often have an enlarged first pair of appendages equipped with claws. Important in the marine food chain, many of the bigger species are of commercial value.

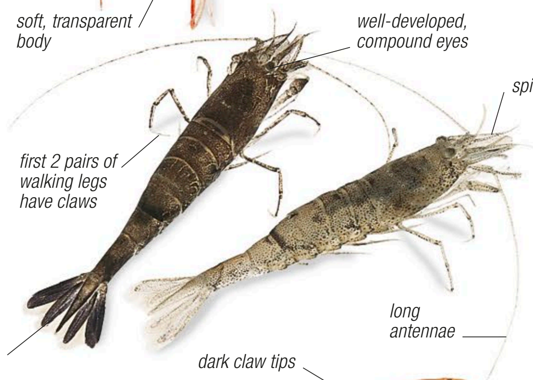
MANTIS SHRIMPS
Mantis shrimps (family Squillidae) use a unique spearing (sometimes smashing) technique to catch prey. The second pair of thoracic legs is large and specially designed for spearing, and is capable of striking very fast indeed. Mantis shrimps have a flattened, flexible body and stalked, compound eyes. Squilla *species, such as the one pictured right, inhabit tropical and subtropical seas, and are territorial.*



KRILL
Krill (family Euphausiidae) are slender, shrimplike crustaceans that filter food particles from sea water using the long hairs on their thoracic legs. They are highly gregarious, and also luminescent. The species shown left, Antarctic krill (Euphausia superba)*, forms very large swarms that sometimes have a collective weight of more than 2 million tons. In the Arctic Ocean, these swarms are an important food for baleen whales.*



PRAWNS
Members of the family Palaemonidae have a carapace that extends forward to form a rostrum, which may have toothed edges. Most species are colorless and transparent. They use their claws to pick up food. The common prawn, Palaemon serratus *(shown left), has very long antennae, up to 1.5 times longer than the body, that warn it of the presence of predators. This species is caught in huge numbers since it is a common human food.*

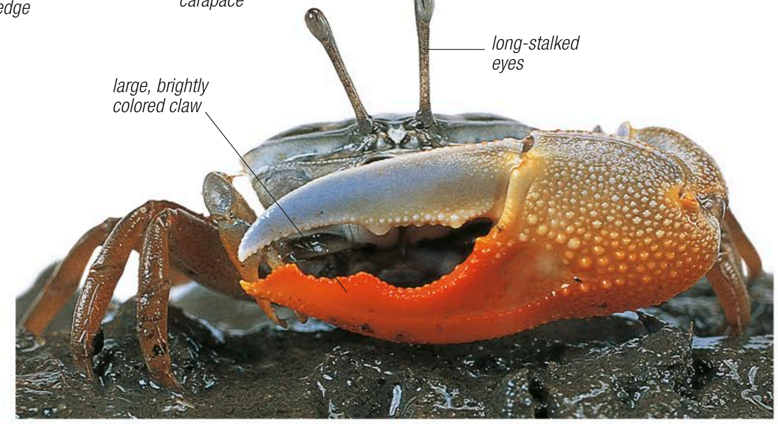


SHRIMPS
Pigment cells (chromatophores) present in their body allow these bottom-dwelling omnivores (family Crangonidae) to change color. The carapace is extended to form a short spinelike, smooth-edged rostrum. The common shrimp, Crangon crangon *(pictured left), is fished commercially along European coastlines.*

BURROWING SHRIMPS
These shrimps (family Callianassidae) make U- or Y-shaped, or branched, networks of burrows in silt and mud in shallow water. They are soft-bodied crustaceans, predatory on small organisms and worms. These shrimps are collected as fish bait in some parts of the world. The picture on the left shows a species from the genus Callianassa. *In this genus, one of the first thoracic legs may be much larger than the other.*

CANCRID CRABS
When adult, cancrid crabs (family Cancridae) typically live on the seabed, where they prey on other invertebrates and scavenge for dead remains. The species shown here is the European edible crab, Cancer pagurus, *which has a distinctive "pie crust" edge to its carapace. Found off the European Atlantic, Mediterranean, and West African coasts, it is up to 12 in (30 cm) across. It is commercially fished in many areas and large specimens are now rare.*

SWIMMING CRABS
Swimming crabs have paddlelike hindlegs that help them to swim efficiently. As in other crabs, the short abdomen is folded like a tail under the body. The common shore crab, Carcinus maenas *(pictured), buries itself in mudflats and sandy regions. Known as the green crab in North America, this species is not as good a swimmer as other members of the family Portunidae.*



GHOST AND FIDDLER CRABS
In the tropics, ghost and fiddler crabs (family Ocypodidae) abound on low-lying coasts. Ghost crabs live in burrows on sandy beaches, and emerge after dark, feeding on debris washed up by the tide. They have keen eyesight, and are extremely rapid runners, disappearing into their burrows at the first hint of danger. Fiddler crabs, which are smaller, burrow into the mud of mangrove swamps. Males have one large claw, which they wave to attract females and to deter rivals. The claw often makes up half the male's weight. Fiddler crabs feed on organic matter present in silt. There are about 97 species; shown here is the male orange fiddler crab, Uca vocans, *from Africa and Southeast Asia.*